

4.1

a. $R^2 = SSR/SST$, $SST = \sum y_i^2 - N\bar{y}^2 = 7825.94 - 20 \cdot 19.21^2 = 445.458$,

$$R^2 = 375.47/445.458 = 0.8429$$

b. $\hat{\sigma}^2 = SSE/N-K$, $0.7911 = 1 - SSE/725.94 \Rightarrow SSE = 151.6489$

$$\hat{\sigma}^2 = 151.6489/18 = 8.4249$$

c. $R^2 = 1 - SSE/SST = 1 - 182.85/631.63 = 0.7105$

4.3

a. $\hat{y}_0 = 1.2 + 0.8 \cdot 4 = 4.4$

b. $\hat{\sigma}^2 = SSE/N-K = 3.6/3 = 1.2$, $se(f) = \sqrt{\hat{\sigma}^2 (1 + 1/N + (x_0 - \bar{x})^2/S_{xx})} \Rightarrow$

$$\sqrt{1.2 \cdot (1 + 0.2 + 3^2/10)} = 1.5875$$

c. 95% P.I. $\Rightarrow [\hat{y}_0 - t_{3,0.025} \cdot se(f), \hat{y}_0 + t_{3,0.025} \cdot se(f)] \Rightarrow 4.4 \pm 3.182 \cdot 1.5875$

$$\Rightarrow [-0.651, 9.451]$$

d. 99% P.I. $\Rightarrow [\hat{y}_0 - t_{3,0.005} \cdot se(f), \hat{y}_0 + t_{3,0.005} \cdot se(f)] \Rightarrow 4.4 \pm 5.841 \cdot 1.5875$

$$\Rightarrow [-4.873, 13.673]$$

e. $\hat{y} = 1.2 + 0.8 \cdot 1 = 2$, $se(f) = \sqrt{1.2 \cdot (1 + 0.2)} = 1.2$

$$95\% \text{ P.I.} \Rightarrow 2 \pm 3.182 \cdot 1.2 \Rightarrow [-1.818, 5.818], \text{ Interval in (c) is larger.}$$

```
Call:
lm(formula = log(price) ~ sqft, data = df)

Residuals:
    Min       1Q   Median       3Q      Max
-1.32528 -0.19199  0.00122  0.21678  0.86609

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  4.393866    0.043288   101.50  <2e-16 ***
sqft         0.036044    0.001486    24.26  <2e-16 ***
```

4.25

a. β_1 is the expected value of $\log(\text{price})$ of the house when $\text{sqft} = 0$. β_2 is the Semi-elasticity. When the house area rises 1 unit, the house price will increase 3.60%.

$$\text{slope} = d\text{PRICE}/d\text{SQFT} = \beta_2 \text{PRICE}$$

$$\text{Elasticity} = (d\text{PRICE}/\text{PRICE})/(d\text{SQFT}/\text{SQFT}) = \beta_2 \text{SQFT}$$

```
> avg_price*beta2
sqft
9.019661
> avg_sqft*beta2
sqft
0.9833701
```

```
Call:
lm(formula = log(price) ~ log(sqft), data = df)

Residuals:
    Min       1Q   Median       3Q      Max
-1.36864 -0.20849  0.00487  0.23204  1.21099

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.04965    0.15797   12.97  <2e-16 ***
log(sqft)    1.02483    0.04839   21.18  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.0
```

- b. α_1 is the expected value of $\log(\text{price})$ of the house when $\log(\text{sqft})=0$. α_2 is the elasticity. When the house area rises 1%, the house price will increase 1.02%.

$$\text{slope} = d\text{PRICE}/d\text{SQFT} = \alpha_2 \text{ PRICE}/\text{SQFT}$$

$$\text{Elasticity} = (d\text{PRICE}/\text{PRICE})/(d\text{SQFT}/\text{SQFT}) = \alpha_2$$

```
> alpha2*avg_price/avg_sqft
log(sqft)
9.399923
> alpha2
log(sqft)
1.024828
```

```
>
> r2_linear
[1] 0.6413167
> r2_gen_a
[1] 0.6621612
> r2_gen_b
[1] 0.6445084
```

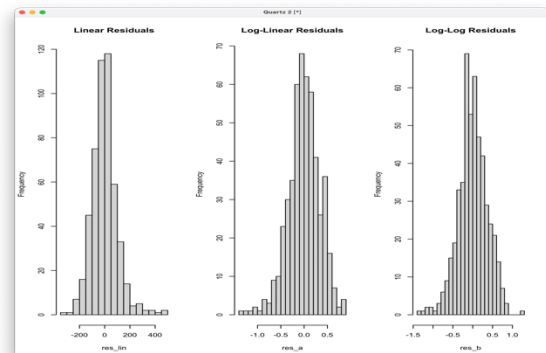
- c. log-linear model has the most explanatory power in this case.

- d. All models are not compatible with normality based on JB test. Linear model even has the right-skew.

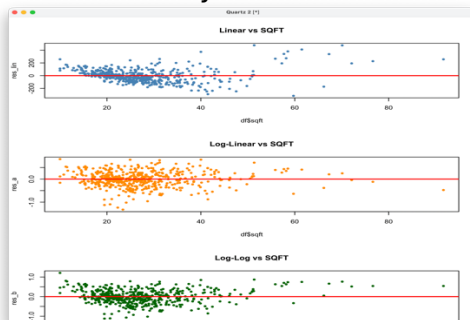
```
$Linear
Jarque Bera Test
data: res_lin
X-squared = 221.11, df = 2, p-value < 2.2e-16

$Log_Linear
Jarque Bera Test
data: res_a
X-squared = 26.679, df = 2, p-value = 1.61e-06

$Log_Log
Jarque Bera Test
data: res_b
X-squared = 14.279, df = 2, p-value = 0.0007933
```



- e. Based on the residual plots, we can conclude that only linear model is not Homoscedasticity due to Funnel-shape.



```
f&g. $Linear      fit      lwr      upr
1 246.4557 44.27727 448.6341

$Log_Linear      fit      lwr      upr
1 214.2336 109.7685 418.1167

$Log_Log      fit      lwr      upr
1 227.5386 111.1406 465.8406
```

- h. Since log-linear model has the greatest R^2 and it is Homoscedasticity. Choose log-linear model.

5.5

$$a. \text{var}(b_2|x) = \sigma^2 / \sum (x_t - \bar{x})^2 = \sigma^2 / \sum t^2 - (\sum t)^2 / T = \sigma^2 / [(T(T+1)(2T+1)/6) - T^2(T^2+1)/4T]$$

$$\Rightarrow 12\sigma^2 / [T(T^2-1)]. \text{ when } T \text{ approaches infinity, limit } \text{var}(b_2|x) = 0$$

$\Rightarrow$ the least squares estimator is consistent.

$$b. \text{var}(b_2|x) = \sigma^2 / \sum (x_t - \bar{x})^2 = \sigma^2 / \sum t^2 - (\sum t)^2 / T = \sigma^2 / (1/3 - 0) = 3\sigma^2.$$

$$\text{when } T \text{ approaches infinity, limit } \text{var}(b_2|x) = 3\sigma^2 \neq 0$$

$\Rightarrow$ the least squares estimator is not consistent.

c. The time trend $x_t = t$ expands infinitely, providing ever-increasing information that drives $\text{var}(b_2|x)$ to zero as $T \rightarrow \infty$. Conversely, the geometric series $x_t = 0.5t$ decays so rapidly that total information is capped by a finite constant, leaving the variance stuck above zero regardless of sample size.

5.6

$$N=63 \quad K=3 \quad df=60 \quad t_{0.025,60}=2$$

$$a. H_0: \beta_2 = 0 \quad H_a: \beta_2 \neq 0, \quad b_2 = 3 \Rightarrow t = 3 - 0 / \sqrt{4} = 1.5$$

$$\text{Since } t = 1.5 < t_{0.025,60} = 2 \Rightarrow \text{Non-reject } H_0$$

$$b. H_0: \beta_1 + 2\beta_2 = 5 \quad H_a: \beta_1 + 2\beta_2 \neq 5, \quad b_1 + 2b_2 = 8 \Rightarrow t = 8 - 5 / \sqrt{3 + 4(-2) + 4 \cdot 4} = 0.904$$

$$\text{Since } t = 0.904 < t_{0.025,60} = 2 \Rightarrow \text{Non-reject } H_0$$

$$c. H_0: \beta_1 - \beta_2 + \beta_3 = 4 \quad H_a: \beta_1 - \beta_2 + \beta_3 \neq 4, \quad b_1 - b_2 + b_3 = -2 \Rightarrow t = -2 - 4 / \sqrt{3 + 4 + 3 - 2(-2 + 0) + 2 \cdot 1} = -1.5$$

$$\text{Since } t = -1.5 < t_{0.025,60} = 2 \Rightarrow \text{Non-reject } H_0$$

5.20

$$a. \text{ apply } y_i = \beta_1 + \beta_2 x_i + e_i, \quad \bar{y}_2 - \bar{y}_1 = (\beta_1 + \beta_2 \bar{x}_2 + \bar{e}_2) - (\beta_1 + \beta_2 \bar{x}_1 + \bar{e}_1) = \beta_2 (\bar{x}_2 - \bar{x}_1) + (\bar{e}_2 - \bar{e}_1), \quad \widehat{\beta_{2,mean}} = \beta_2 + \frac{\bar{e}_2 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1},$$

$$E(\widehat{\beta_{2,mean}}|\mathbf{x}) = \beta_2 + \frac{E(\bar{e}_2|\mathbf{x}) - E(\bar{e}_1|\mathbf{x})}{\bar{x}_2 - \bar{x}_1}, \text{ By MR3 } E(e_i) = 0$$

$$\Rightarrow E(\bar{e}_2) = 0 \ \& \ E(\bar{e}_1) = 0$$

$$\text{So, } E(\widehat{\beta_{2,mean}}|x) = \beta_2$$

$$\text{b. } \text{var}(\widehat{\beta_{2,mean}}|\mathbf{x}) = \text{var}\left(\beta_2 + \frac{\bar{e}_2 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1} \middle| \mathbf{x}\right) = \frac{\text{var}(\bar{e}_2 - \bar{e}_1|\mathbf{x})}{(\bar{x}_2 - \bar{x}_1)^2}$$

Since they have the first half and the second half sample

$$\Rightarrow \text{var}(\bar{e}_2 - \bar{e}_1) = \text{var}(\bar{e}_2) + \text{var}(\bar{e}_1) = \frac{4\sigma^2}{N}$$

$$\Rightarrow \text{var}(\widehat{\beta_{2,mean}}|\mathbf{x}) = \frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}$$

$$\text{c. } \text{var}(\widehat{\beta_{2,mean}}) = E[\text{var}(\widehat{\beta_{2,mean}}|\mathbf{x})] + \text{var}(E[\widehat{\beta_{2,mean}}|\mathbf{x}])$$

$$\text{Since } E[\widehat{\beta_{2,mean}}|\mathbf{x}] = \beta_2 \Rightarrow \text{var}(\beta_2) = 0$$

$$\Rightarrow \text{var}(\widehat{\beta_{2,mean}}) = E[\text{var}(\widehat{\beta_{2,mean}}|\mathbf{x})] = \frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}$$

d. An estimator is consistent if its variance collapses to zero as the sample size

$$N \text{ approaches infinity: } \lim_{N \rightarrow \infty} \text{var}(\widehat{\beta_{2,mean}}|\mathbf{x}) = 0 \Rightarrow \lim_{N \rightarrow \infty} \frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2} = 0$$

and $(\bar{x}_2 - \bar{x}_1)^2$ not converge to 0.

$$\text{e. } |N| \quad \text{var}(b_2|x) \quad |\text{var}(\widehat{\beta_{2,mean}}|\mathbf{x})| \quad |E[(s_x^2)^{-1}]| \quad |E\left[\frac{4}{(\bar{x}_2 - \bar{x}_1)^2}\right]|$$

`print(results)`

	N	var_ols	var_mean	inv_sx2	term_mean
	100	1.17415057	1.44588776	0.1174151	0.1445888
	500	0.24703852	0.33081080	0.1235193	0.1654054
	1000	0.12070993	0.15939763	0.1207099	0.1593976
	5000	0.02422763	0.03238291	0.1211382	0.1619146

f. The results confirm that the OLS estimator is more efficient than the mean-

based estimator, as $\text{var}(b_2|x) < \text{var}(\widehat{\beta_{2,mean}}|\mathbf{x})$ across all N, consistent with

the **Gauss-Markov Theorem**. Both estimators are **consistent**, with their

variances approaching zero as N increases. However, the mean-based estimator

converges to a higher limit (0.16) than OLS (0.12), representing a permanent efficiency loss of approximately **25%** regardless of sample size.

g. Given $x \sim U(0,10)$, the population variance is $\sigma_x^2 = \frac{100}{12} = 8.333$

By the Law of Large Numbers: $(s_x^2)^{-1} \rightarrow \frac{1}{\sigma_x^2} = \frac{12}{100} = \mathbf{0.12}$.

And $\bar{x}_1 = 2.5$, $\bar{x}_2 = 7.5$, $\frac{4}{(\bar{x}_2 - \bar{x}_1)^2} = 0.16$

h. If observations are randomly assigned instead of ordered by x , the estimator **will not be consistent**.

Without ordering, both $\bar{x}_1$ and $\bar{x}_2$ are just random samples from the same distribution. As N increases, both group means will converge to the same population mean, causing their difference $(\bar{x}_2 - \bar{x}_1)$ to approach **zero**. In the variance formula, having a denominator that approaches zero prevents the overall variance from shrinking to zero as $N \rightarrow \infty$. Therefore, ordering is a **strict requirement** to ensure the estimator provides a reliable and consistent result.

